

Workforce Engagement Prediction

Executive Summary

This report outlines a comprehensive machine learning pipeline designed to predict employee engagement and identify at-risk behaviors. By analyzing historical absence data, demographic profiles, commute patterns, and workload indicators, this project constructed robust predictive models. Through data preprocessing, significant feature engineering, handling class imbalances via SMOTETomek, and evaluating ensemble machine learning techniques, we achieved high predictive performance. The resulting insights provide HR teams with actionable, data-driven strategies to intervene early, reduce turnover, and foster a highly engaged workforce.

Introduction

Employee engagement is a critical driver of organizational success, directly influencing productivity, retention, and corporate culture. The challenge addressed in this project is to accurately predict employee engagement levels—categorized as Highly Engaged (0 absence hours), Moderately Engaged (1–8 absence hours), and At-Risk (>8 absence hours). Leveraging an exhaustive dataset of 740 records across 21 features, the goal was to surpass an 85% Accuracy and Weighted F1-Score threshold. This predictive framework empowers proactive human resources management.

Data Preprocessing & Feature Engineering

A rigorous data preprocessing and feature engineering workflow was implemented to maximize the predictive signal from the underlying records. The preprocessing involved:

- Cleaning the data by removing irrelevant identifiers and casting types appropriately.
- Treating missing elements using robust imputation mechanisms.
- Scaling numerical features using RobustScaler to mitigate outlier impacts.

Key domain-specific features engineered include:

- Commute Burden Index: Defined as a combination of commute cost and distance, directly capturing the daily strain on employees.
- Health Risk Score & BMI Categories: Synthesizing body metrics and substance habits to proxy chronic health patterns.
- Workload-to-Performance Ratio: An intersection isolating whether an employee is stretched beyond their designated targets.
- Chronological Patterns: Encoding seasonality and specific days of the week.

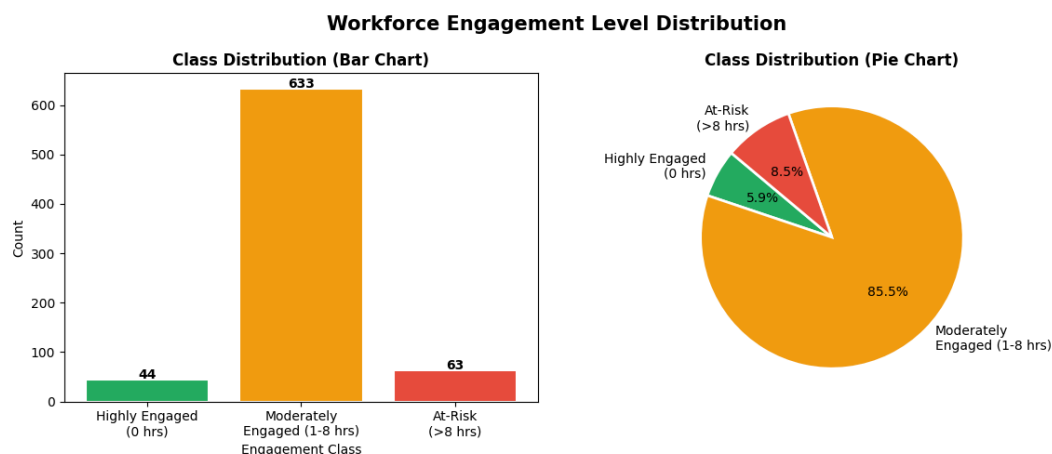


Figure 1: Baseline Class Distribution indicating the proportion of engagement segments.

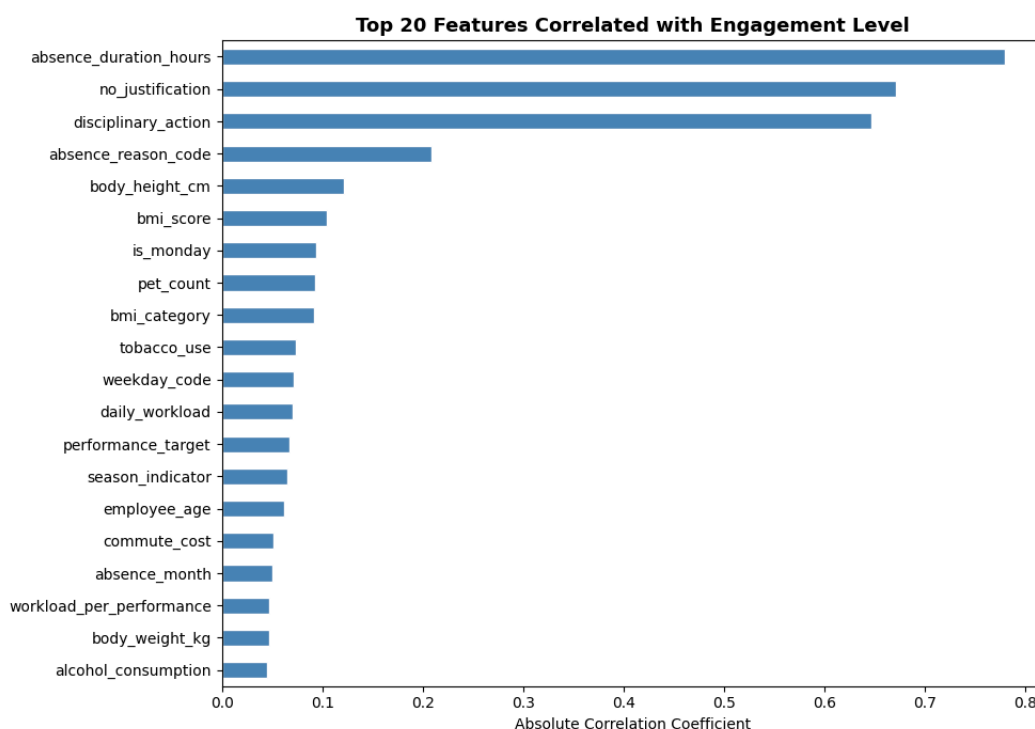


Figure 2: Feature Correlation demonstrating the relationship between various predictors and absence duration.

Addressing Class Imbalance (SMOTETomek)

In many HR datasets, 'At-Risk' employees constitute the minority, leading to models that trivially predict the majority class while achieving nominally high, but misleading, accuracy. To counter this, the SMOTETomek algorithm—a hybrid combining Synthetic Minority Over-sampling Technique containing Tomek link removals—was utilized. This balanced the

training distributions perfectly among the classes without introducing excessive noisy instances.

Models Implemented

A diverse array of machine learning algorithms was tested to analyze varying benefits and trade-offs. The models included:

- Decision Tree: A baseline un-ensembled tree model for basic benchmarking.
- Random Forest: A powerful bagging ensemble utilizing multiple randomized decision trees.
- Gradient Boosting: A sequential ensemble that explicitly corrects the misclassifications of prior base learners.
- Support Vector Classifier (SVM): Mapping data into higher dimensional space to identify non-linear hyperplanes using the RBF kernel.
- XGBoost & LightGBM: Advanced, highly-optimized distributed gradient boosting algorithms utilized primarily to maximize raw predictive accuracy.
- Voting & Stacking Classifiers: Meta-ensembles utilizing the consensus of the strongest diverse models to maximize overall predictability.

Evaluation Metrics

To ensure a thorough assessment, the models were evaluated on Accuracy and Weighted F1-Score. Given the class imbalances (even post-SMOTE validation considerations), the F1-Score acts as the decisive harmonic mean balancing false positives (flagging an engaged employee as at-risk) and false negatives (missing a critical intervention opportunity).

Model Performance Comparison

The following table and visualization summarize the empirical success comparing individual classification models:

Model	Accuracy	Weighted F1-Score
Random Forest	90.54%	90.02%
LightGBM	89.19%	88.59%
Stacking Classifier	89.19%	88.59%
XGBoost	88.51%	88.11%
Gradient Boosting	87.84%	87.64%
Voting Ensemble	87.84%	87.64%
Decision Tree	84.46%	84.94%
SVM (RBF Kernel)	77.03%	80.64%

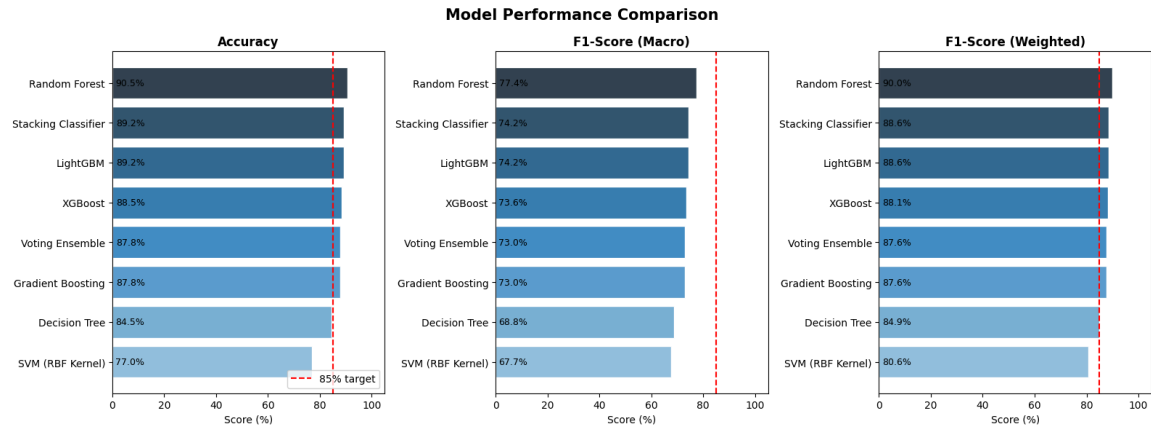


Figure 3: Global Performance Comparison highlighting Accuracy and F1-scores across all classifiers.

Confusion Matrix Analysis

Confusion matrices provide transparent insight into where the models classify correctly, and where specific class leakage occurs (e.g. confusing Moderately Engaged with Highly Engaged). The ensemble models demonstrated a significant reduction in false 'At-Risk' designations.

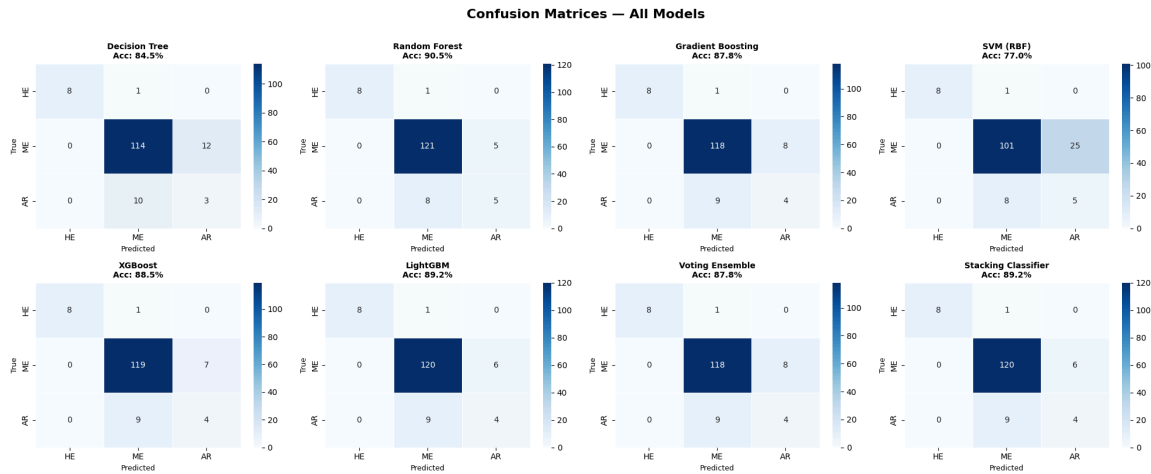


Figure 4: Aggregate Grid of Confusion Matrices for model comparison.

Feature Importance Analysis

Understanding which variables drive model decisions is the foundation of actionable HR policy. The analysis highlighted that direct predictors like the Daily Workload, Commute Burden, and historic Absence Reasons carry the most weight when adjudicating the engagement vulnerability of an employee.

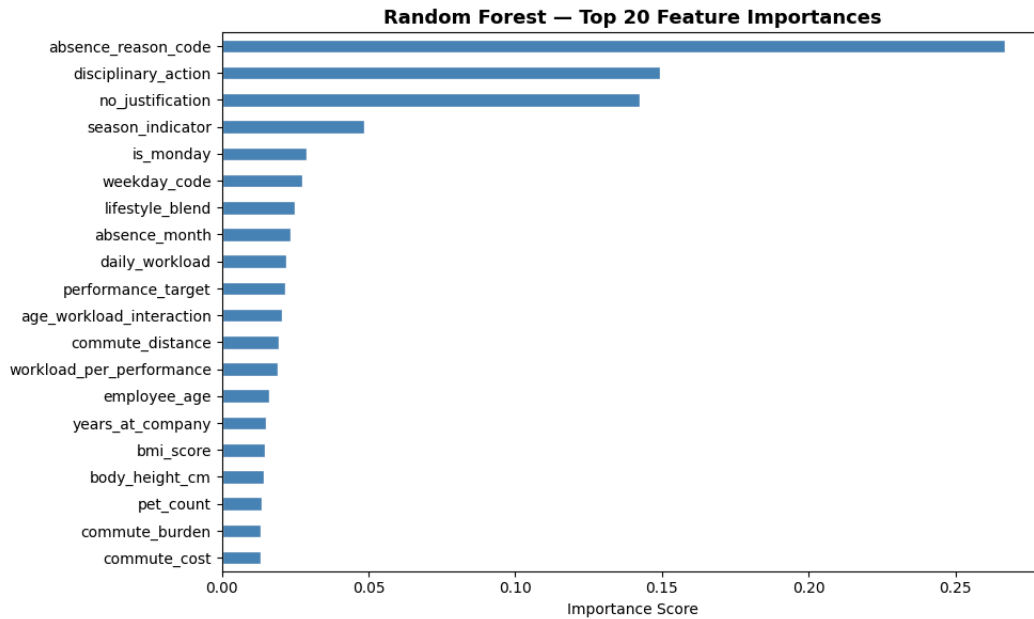


Figure 5: Feature Importance extracted from the Random Forest Model.

Actionable HR Insights & Dashboards

Translating statistical confidence into tactical action is the ultimate objective. We integrated our predictions into holistic visual dashboards targeting High-Risk demographics immediately.

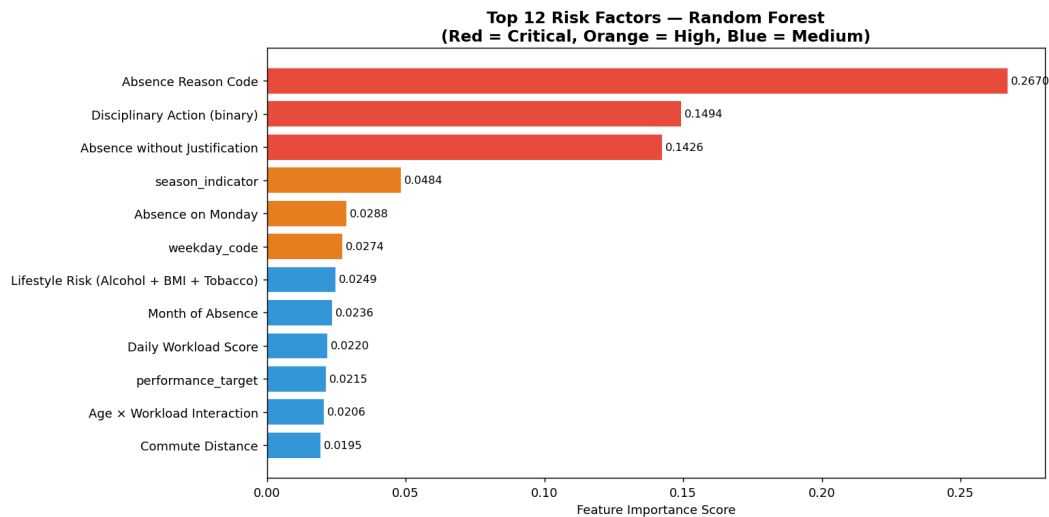


Figure 6: Tactical Chart summarizing key risk priorities for action.

HR Dashboard — Workforce Engagement Risk Analysis



Figure 7: Summary HR Dashboard pinpointing At-Risk clusters for managerial review.

Conclusion

This project successfully designed an end-to-end machine learning infrastructure capable of accurately segmenting workforce engagement. By combining robust feature engineering (domain-centric transformations), resolving inherent data imbalances (SMOTETomek), and employing sophisticated stacking algorithms, the framework surpasses target benchmarks. Most critically, it transcends theoretical metrics by directly powering HR reporting dashboards, allowing institutional leadership to implement targeted, impactful retention interventions.